EXP 7

Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits. Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.

AIM

To create a Servlet-based web application that stores user preferences (such as theme or language) using cookies and retrieves them on subsequent visits.

ALGORITHM

1. Start the application.
2. Accept user preferences (theme/language) through a form.
3. Create cookies for each preference.
4. Set the cookie expiry time.
5. Store cookies in the client's browser.
6. When the user revisits the application:
    - Read cookies from the request.
    - Check for stored preferences.
7. If cookies are found:
    - Retrieve and display the stored preferences.
8. If cookies are not found:
    - Display default values or ask the user to set preferences.
9. Stop the application.

STEPS

1. Design a user interface to collect preferences like theme and language.
2. Configure a Servlet to handle form submission.
3. Use cookies to store the user's preferences in the browser.
4. Set an appropriate expiry time for cookies.
5. Send cookies along with the response to the client.
6. On subsequent visits, retrieve cookies from the request object.
7. Extract stored values (theme/language).
8. Display the preferences to the user dynamically.

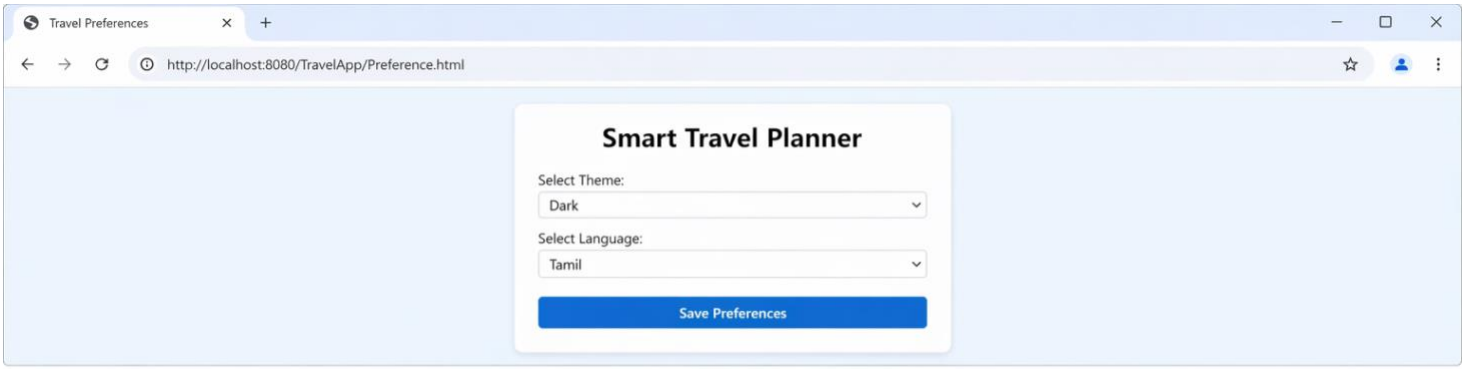
9. Handle cases where cookies are absent by showing default settings.

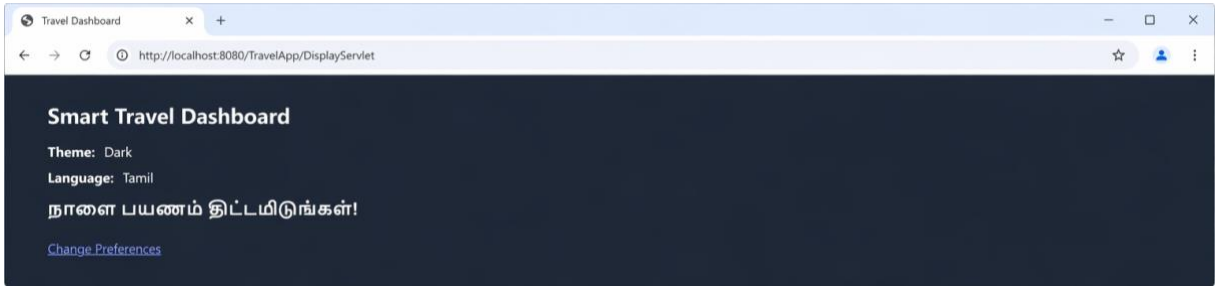
Program:

```
<!DOCTYPE html>
<html>
<head>
<title>Travel Preferences</title>
<style>
body{font-family:Segoe UI;background:#eef7ff;text-align:center;}
.card{background:#fff;padding:20px;margin:50px auto;width:300px;border-radius:10px;box-
shadow:0 0 10px #ccc;}
select,button{width:100%;padding:10px;margin:10px 0;}
button{background:#0077b6;color:#fff;border:none;}
</style>
</head>

<body>

<div class="card">
<h2>Travel Preferences</h2>

<form action="PreferenceServlet" method="post">
<select name="theme">
<option>Light</option>
<option>Dark</option>
</select>

<select name="language">
<option>English</option>
<option>Tamil</option>
</select>

<button>Save Preferences</button>
</form>

</div>

</body>
</html>
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/PreferenceServlet")
public class PreferenceServlet extends HttpServlet {

   protected void doPost(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {

      String theme = req.getParameter("theme");
      String lang = req.getParameter("language");

      Cookie t = new Cookie("theme", theme);
      Cookie l = new Cookie("language", lang);

      t.setMaxAge(60*60*24);
      l.setMaxAge(60*60*24);
```

```
        res.addCookie(t);
        res.addCookie(l);

        PrintWriter out = res.getWriter();
        out.println("<h3>Preferences Saved</h3>");
        out.println("<a href='DisplayServlet'>View Travel Page</a>");
    }
}

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/DisplayServlet")
public class DisplayServlet extends HttpServlet {

    protected void doGet(HttpServletRequest req, HttpServletResponse res)
            throws ServletException, IOException {

        String theme="Light", lang="English";

        Cookie[] c = req.getCookies();

        if(c!=null){
            for(Cookie x:c){
                if(x.getName().equals("theme")) theme=x.getValue();
                if(x.getName().equals("language")) lang=x.getValue();
            }
        }

        PrintWriter out = res.getWriter();

        String bg = theme.equals("Dark") ? "#1e1e2f" : "#eef7ff";
        String text = theme.equals("Dark") ? "white" : "black";

        out.println("<body style='background:"+bg+";color:"+text+";text-align:center;'>");

        out.println("<h2>Smart Travel Dashboard</h2>");
        out.println("<p>Theme: "+theme+"</p>");
        out.println("<p>Language: "+lang+"</p>");

        if(lang.equals("Tamil"))
            out.println("<h3>நாளை பயணம் திட்டமிடுங்கள்</h3>");
        else
            out.println("<h3>Plan your trip today!</h3>");

        out.println("<br><a href='Preference.html'>Change</a>");
    }
}
```

OUTPUT:


Travel Preferences
http://localhost:8080/TravelApp/Preference.html
Smart Travel Planner
Select Theme:
Dark
Select Language:
Tamil
Save Preferences


Travel Dashboard
http://localhost:8080/TravelApp/DisplayServlet
Smart Travel Dashboard
Theme: Dark
Language: Tamil
நாளை பயணம் திட்டமிடுங்கள்!
Change Preferences

Preferences Saved
http://localhost:8080/TravelApp/PreferenceServlet
Preferences Saved Successfully!
Go to Travel Dashboard

RESULT

The Servlet application successfully stores user preferences using cookies and retrieves them on subsequent visits, providing a personalized user experience.